Homework 6 - Part 1b: 1D Wedge Bearing Verification

Problem Setup

To verify the finite element solver against the analytical solution for a 1D wedge bearing, the physical parameters were adjusted to represent a pure wedge. Specifically, the leading taper angle was set to zero ($T_a = 0$), and Neumann boundary conditions were enforced on the top and bottom edges of the domain. This was achieved by applying Dirichlet boundary conditions ($p = 0$) exclusively to the leading ($x = 0$) and trailing ($x = L$) edges, thus forcing a strictly 1D flow in the x-direction.

The analytical solution for the 1D wedge bearing is given by:

$$\tilde{p}(x) = \frac{\alpha\Lambda}{1-\alpha^2}\left(\frac{1}{H^2} - \frac{1}{\alpha^2}\right) - \frac{\Lambda}{1-\alpha}\left(\frac{1}{H} - \frac{1}{\alpha}\right) \tag{1}$$

where $\alpha = h_1/h_2$, $H(x) = \alpha + (1-\alpha)(x/L)$, and $\Lambda = 6\mu UL/h_2^2$.

Convergence Study Results

A mesh refinement study was performed by varying the number of elements in the x-direction ($E_x \in \{15, 30, 60, 120\}$) while keeping $E_y = 40$ constant. The maximum pointwise error between the numerical solution $p(x)$ and the analytical solution $\tilde{p}(x)$ was computed for each mesh resolution.

Number of Elements (E_x)	**Max Pointwise Error**
15	2932.079
30	757.7912
60	201.5704
120	56.73146

Table 1: Maximum pointwise error in $p(x)$ for different mesh resolutions.

The estimated convergence rate, calculated from the slope of the log-log plot of the error versus E_x, is approximately -1.90. This value is close to -2.0, successfully demonstrating the expected second-order convergence of the linear triangular elements for this problem.

Figures

Figure 1 shows the pressure distribution, while Figure 2 illustrates the convergence behavior of the finite element solver.

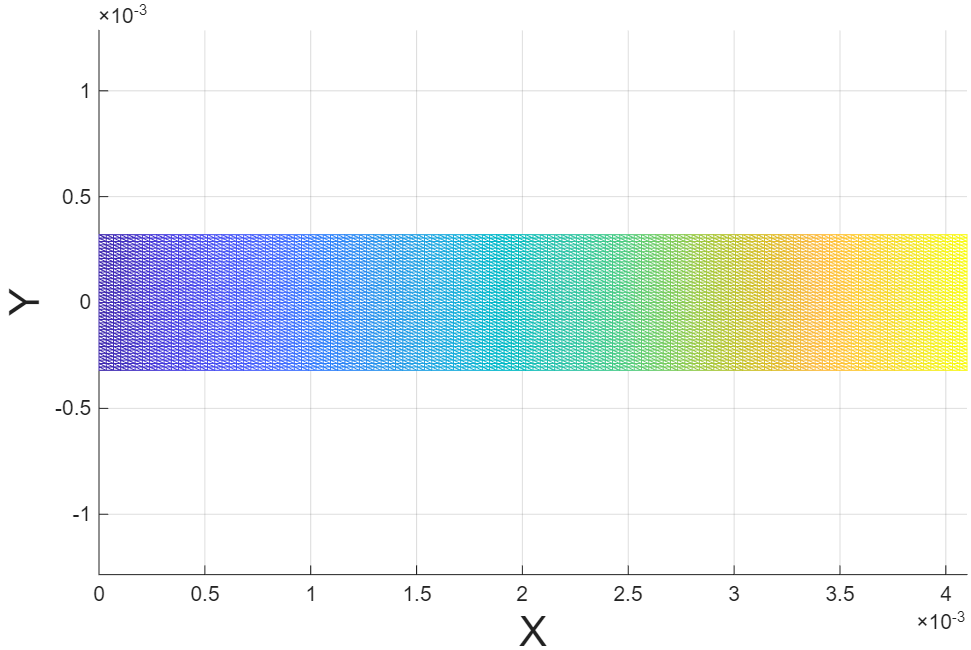

Figure 1: Numerical pressure profile for the 1D wedge bearing setup.

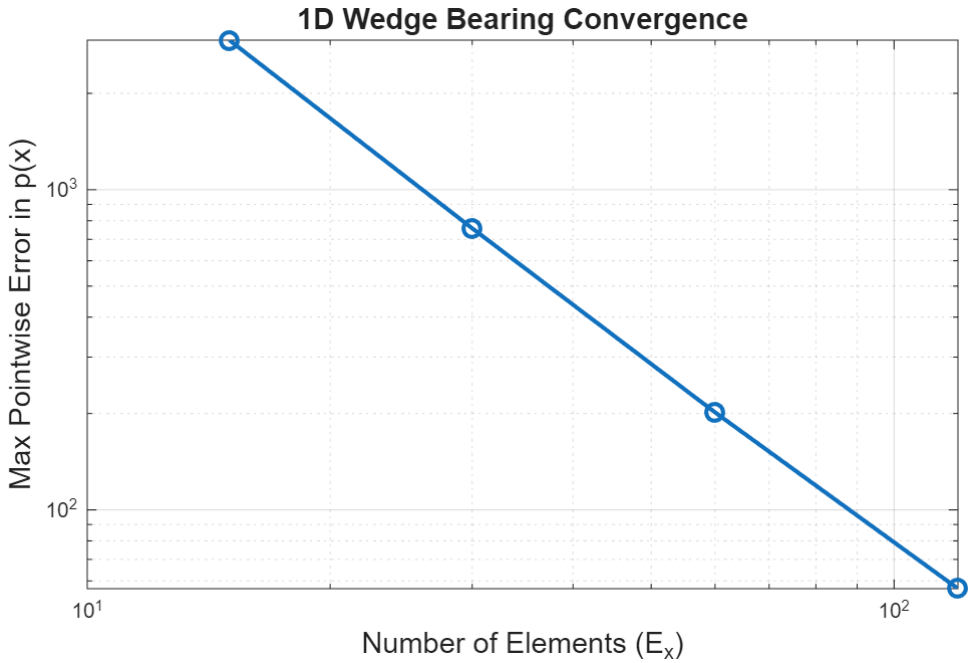

Figure 2: Log-log plot of the maximum pointwise error versus the number of elements (E_x), demonstrating near second-order convergence with a slope of -1.90.